Domenico De Giorgio

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Portfolio | DomDegi | Domenico De Giorgio

Education

Politecnico di Milano M.Sc. High-Performance Computing Engineering — Sep 2025 – Present
Parallel Computing, Advanced Computer Architectures, Numerical Linear Algebra, Partial Differential Equations

Politecnico di Milano B.Sc. Engineering of Computing Systems — Sep 2022 – Sep 2025
Data Structures & Algorithms, Operating Systems, Probability & Statistics, Operations Research

Selected Projects

Adaptive Space-Time Finite Element Solver — Dec 2025 – Jan 2026
- Engineered a distributed parallel solver for heat equation PDEs using the **deal.II** library and **MPI**.
- Implemented space-time adaptive mesh refinement to optimize computational load across distributed nodes, significantly reducing execution time for complex finite element simulations.
- *Stack*: **C++**, **MPI**, **deal.II**, **HPC**, **Distributed Systems**.

Advanced Monte Carlo Integration & Optimization — Dec 2025 – Jan 2026
- Built a high-performance, multithreaded library for N-dimensional Monte Carlo integration and MCMC.
- Leveraged **OpenMP** for parallelization, heavily optimizing cache locality to maximize **CPU core utilization**.
- *Stack*: **C++**, **OpenMP**, **Stochastic Optimization**, **Cache Optimization**.

Client-Server Multiplayer Game — Feb 2025 – Jun 2025
- Collaborated in a 4-person team to architect the complete **Model** and **Controller** layers (**MVC pattern**) for a real-time multiplayer digital board game.
- Managed state synchronization and multi-threading through **RMI** and **TCP/IP Sockets**, achieving 97% **JUnit** test coverage.
- Implemented a Graphical User Interface using **JavaFX** and **SceneBuilder**.
- *Stack*: **Java**, **Sockets**, **MVC**, **Maven**, **JUnit**, **JavaFX**.

High-Performance CUDA Batched Matrix Multiplication — Oct 2026
- Developed a highly optimized custom **CUDA** kernel to accelerate batched matrix multiplications.
- Maximized memory bandwidth and computational throughput by implementing **shared memory tiling**, **thread coarsening**, and **bank conflict avoidance** strategies.
- *Stack*: **C/C++**, **CUDA**, **GPU Computing**, **Memory Profiling**.

Declarative Ephemeral NixOS System – Personal Daily Driver — Apr 2026 – Present
- Architected and maintained a fully reproducible, ephemeral **Linux** environment utilizing the “erase your darlings” (**tmpfs** on root) methodology.
- Managed system packages and user dotfiles declaratively via **Nix Flakes** and **Home Manager**, enabling zero-touch provisioning and version-controlled system states.
- *Stack*: **Nix**, **NixOS**, **Home Manager**, **Linux SysAdmin**, **Git**.

Skills

Languages	C, C++, Python, Java, Nix, Bash, SQL
HPC & Parallel	CUDA, MPI, OpenMP, GPU Computing, Cluster environments
Systems & Tools	Linux (Ubuntu, Arch, NixOS declarative config), Git, Docker, CMake, Slurm
ML & Math	PyTorch, Eigen, deal.II, Numerical Analysis, Pandas, Polars
Spoken	Italian (Native), English (C1 Equivalent – TOEIC: 985/990)